

SAINSHAND, Mongolia

WMO: 443540

Lat: 44.868N

Lon: 110.117E

Elev: 938

StdP: 90.55

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.2	-25.2	-31.9	0.2	-24.2	-29.8	0.3	-23.4	15.6	-15.3	13.6	-16.0	3.5	320	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	34.7	18.9	32.9	18.2	31.0	17.4	21.1	31.0	19.8	29.0	18.8	27.6	4.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	14.7	25.1	16.9	13.5	23.7	15.7	12.5	23.2	66.1	30.9	61.4	28.8	57.8	27.8	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.1	10.4	-29.3	37.8	2.9	1.6	-31.4	38.9	-33.1	39.9	-34.7	40.8	-36.8	41.9		
(5)			-29.3	22.6	3.1	1.7	-31.5	23.8	-33.3	24.8	-35.1	25.7	-37.3	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-16.9	-11.4	-1.8	7.8	15.1	21.5	24.6	22.3	15.7	6.0	-5.9	-14.5		
	DBStd	15.16	4.68	5.49	5.80	5.31	5.03	3.87	3.21	3.40	4.56	4.99	6.03	4.67		
	HDD10.0	3304	835	598	367	103	13	0	0	0	8	142	477	761		
	HDD18.3	5243	1093	832	624	316	126	14	1	6	102	382	727	1019		
	CDD10.0	1583	0	0	1	37	169	344	452	381	180	18	0	0		
	CDD18.3	480	0	0	0	1	25	108	194	129	23	0	0	0		
	CDH23.3	5111	0	0	0	22	348	1171	2108	1235	224	3	0	0		
	CDH26.7	2059	0	0	0	3	113	472	941	474	55	1	0	0		
(14) Wind	WSAvg	4.5	4.0	4.2	5.0	5.5	5.8	4.7	4.1	3.7	4.1	4.3	4.1	4.2		
(15) Precipitation	PrecAvg	110	1	1	2	2	8	18	30	28	11	4	2	2		
	PrecMax	229	6	11	11	15	32	74	136	104	40	26	14	40		
	PrecMin	64	0	0	0	0	0	0	0	2	0	0	0	0		
	PrecStd	42	1	2	3	3	7	16	23	22	10	6	3	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.1	5.6	17.7	26.1	33.2	36.2	38.0	35.9	30.8	22.5	11.5	-0.1		
		MCWB	-4.5	0.1	6.5	12.2	15.6	18.6	20.5	18.8	16.6	11.7	4.3	-3.1		
	2%	DB	-5.4	2.5	13.5	22.9	29.7	33.5	35.3	33.2	28.2	19.2	7.9	-3.3		
		MCWB	-7.8	-2.0	4.2	10.2	13.9	17.3	19.8	18.6	15.1	9.5	2.2	-6.0		
	5%	DB	-7.7	-0.2	10.4	20.0	27.2	31.3	33.4	31.2	25.9	16.5	5.3	-5.8		
		MCWB	-9.7	-3.7	2.8	8.7	13.0	17.0	19.0	17.8	14.2	7.9	0.4	-7.9		
	10%	DB	-9.9	-2.7	7.5	17.2	24.7	29.4	31.5	29.2	23.6	14.1	2.7	-7.8		
		MCWB	-11.4	-5.6	0.9	7.4	11.8	16.0	18.1	16.8	13.1	6.5	-1.3	-9.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.3	0.9	6.9	13.5	17.0	20.7	23.6	22.2	18.7	12.7	4.8	-3.3		
		MCDB	-1.0	4.8	16.7	23.9	30.1	32.6	35.1	31.1	27.1	20.9	10.7	-0.2		
	2%	WB	-7.6	-1.7	4.9	10.7	15.0	19.3	21.8	20.6	16.7	10.1	2.3	-5.8		
		MCDB	-5.4	2.6	13.2	21.4	26.9	30.3	32.0	29.0	24.6	18.3	7.8	-3.4		
	5%	WB	-9.6	-3.6	3.0	9.2	13.7	18.1	20.4	19.3	15.2	8.3	0.6	-7.7		
		MCDB	-7.7	-0.1	10.3	19.3	25.1	28.0	30.1	27.3	23.4	15.5	5.3	-5.8		
	10%	WB	-11.3	-5.4	1.1	7.6	12.3	17.1	19.3	18.2	13.9	6.8	-1.0	-9.3		
		MCDB	-9.8	-2.4	7.5	17.0	23.3	26.6	28.2	26.4	22.1	13.8	2.8	-7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	11.8	12.8	13.1	12.9	11.7	10.9	11.0	11.9	11.6	10.2	8.9		
		MCDBR	11.7	14.3	15.9	16.5	16.1	14.1	13.2	13.2	14.4	14.1	12.8	10.7		
		MCWBR	9.7	10.8	10.0	9.7	8.5	6.5	5.7	5.8	7.1	8.4	8.8	8.6		
		MCDBR	11.5	14.0	15.5	15.4	14.6	12.3	11.4	11.5	12.2	13.4	12.6	10.6		
	5% WB	MCWBR	9.6	10.6	10.0	9.5	8.5	6.3	6.0	6.2	7.2	8.5	8.9	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.285	0.323	0.377	0.421	0.450	0.419	0.397	0.362	0.316	0.276	0.259		
	taud		2.324	2.263	2.197	2.086	2.008	2.017	2.173	2.209	2.237	2.303	2.327	2.306		
	Ebn at Noon		863	900	905	879	849	823	846	852	853	849	832	822		
	Edh at Noon		77	100	124	152	170	169	143	133	119	95	77	71		
(44) All-Sky Solar Radiation	RadAvg		2.07	3.21	4.87	6.17	6.92	6.88	6.64	6.07	5.03	3.68	2.44	1.76		
	RadStd		0.23	0.23	0.22	0.28	0.32	0.37	0.28	0.20	0.20	0.15	0.14	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.33	N/A	N/A	N/A	-0.85	-0.82	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	-1.16	N/A	-0.62	-0.78	N/A	N/A	N/A	N/A

Nomenclature: See separate page